

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
import pandas as pd

url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv"

df = pd.read_csv(url)

df.head()
```

|   | PassengerId | Survived | Pclass | Name                                              | Sex    | Age  | SibSp | Parch | Ticket           | Fare    | Cabin | Embarked |
|---|-------------|----------|--------|---------------------------------------------------|--------|------|-------|-------|------------------|---------|-------|----------|
| 0 | 1           | 0        | 3      | Braund, Mr. Owen Harris                           | male   | 22.0 | 1     | 0     | A/5 21171        | 7.2500  | NaN   | S        |
| 1 | 2           | 1        | 1      | Cumings, Mrs. John Bradley (Florence Briggs Th... | female | 38.0 | 1     | 0     | PC 17599         | 71.2833 | C85   | C        |
| 2 | 3           | 1        | 3      | Heikkinen, Miss. Laina                            | female | 26.0 | 0     | 0     | STON/O2. 3101282 | 7.9250  | NaN   | S        |

```
import os

base_path = "/content/drive/MyDrive/DataScienceLab"

print(os.listdir(base_path))

['Dataset', '.ipynb_checkpoints', 'Output', 'Notebook', 'Screenshots']
```

```
DATASET_PATH = "/content/drive/MyDrive/DataScienceLab/Dataset"
OUTPUT_PATH = "/content/drive/MyDrive/DataScienceLab/Output"
SCREENSHOT_PATH = "/content/drive/MyDrive/DataScienceLab/Screenshots"
```

```
import pandas as pd

url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv"

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```

```
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|   | PassengerId | Survived | Pclass | Name                                              | Sex    | Age  | SibSp | Parch | Ticket           | Fare    | Cabin | Embarked |
|---|-------------|----------|--------|---------------------------------------------------|--------|------|-------|-------|------------------|---------|-------|----------|
| 0 | 1           | 0        | 3      | Braund, Mr. Owen Harris                           | male   | 22.0 | 1     | 0     | A/5 21171        | 7.2500  | NaN   | S        |
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```
df.to_csv(
    f"{DATASET_PATH}/Titanic_Original.csv",
    index=False
)

print("Dataset Saved")
```

Dataset Saved

```
print(df.shape)
```

(891, 12)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   PassengerId  891 non-null    int64
1   Survived    891 non-null    int64
```

```

2  Pclass      891 non-null   int64
3  Name        891 non-null   object
4  Sex         891 non-null   object
5  Age         714 non-null   float64
6  SibSp       891 non-null   int64
7  Parch       891 non-null   int64
8  Ticket      891 non-null   object
9  Fare        891 non-null   float64
10 Cabin       204 non-null   object
11 Embarked    889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

```

```
df.describe()
```

|              | PassengerId | Survived   | Pclass     | Age        | SibSp      | Parch      | Fare       |
|--------------|-------------|------------|------------|------------|------------|------------|------------|
| <b>count</b> | 891.000000  | 891.000000 | 891.000000 | 714.000000 | 891.000000 | 891.000000 | 891.000000 |
| <b>mean</b>  | 446.000000  | 0.383838   | 2.308642   | 29.699118  | 0.523008   | 0.381594   | 32.204208  |
| <b>std</b>   | 257.353842  | 0.486592   | 0.836071   | 14.526497  | 1.102743   | 0.806057   | 49.693429  |
| <b>min</b>   | 1.000000    | 0.000000   | 1.000000   | 0.420000   | 0.000000   | 0.000000   | 0.000000   |
| <b>25%</b>   | 223.500000  | 0.000000   | 2.000000   | 20.125000  | 0.000000   | 0.000000   | 7.910400   |
| <b>50%</b>   | 446.000000  | 0.000000   | 3.000000   | 28.000000  | 0.000000   | 0.000000   | 14.454200  |
| <b>75%</b>   | 668.500000  | 1.000000   | 3.000000   | 38.000000  | 1.000000   | 0.000000   | 31.000000  |
| <b>max</b>   | 891.000000  | 1.000000   | 3.000000   | 80.000000  | 8.000000   | 6.000000   | 512.329200 |

```
df.isnull().sum()
```

|                    | 0   |
|--------------------|-----|
| <b>PassengerId</b> | 0   |
| <b>Survived</b>    | 0   |
| <b>Pclass</b>      | 0   |
| <b>Name</b>        | 0   |
| <b>Sex</b>         | 0   |
| <b>Age</b>         | 177 |
| <b>SibSp</b>       | 0   |
| <b>Parch</b>       | 0   |
| <b>Ticket</b>      | 0   |
| <b>Fare</b>        | 0   |
| <b>Cabin</b>       | 687 |
| <b>Embarked</b>    | 2   |

```
dtype: int64
```

```

missing = df.isnull().sum()

missing.to_csv(
    f"{OUTPUT_PATH}/Missing_Report.csv"
)

```

```

df['Age'].fillna(
    df['Age'].median(),
    inplace=True
)

```

/tmp/ipykernel\_1125/1348529174.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through c  
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are  
For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]

```
df['Age'].fillna(
```

```

df.drop(
    columns=['Cabin'],
    inplace=True
)

```

```
df.isnull().sum()
```

```

      0
PassengerId  0
Survived     0
Pclass       0
Name         0
Sex          0
Age         0
SibSp        0
Parch        0
Ticket       0
Fare         0
Embarked     2

```

```
dtype: int64
```

```

print(
    "Duplicates:",
    df.duplicated().sum()
)

```

```
Duplicates: 0
```

```
df.drop_duplicates(inplace=True)
```

```
df.select_dtypes(include='object').columns
```

```
Index(['Name', 'Sex', 'Ticket', 'Embarked'], dtype='object')
```

```

from sklearn.preprocessing import LabelEncoder

encoder = LabelEncoder()

df['Sex'] = encoder.fit_transform(
    df['Sex']
)

```

```

df['Embarked'] = encoder.fit_transform(
    df['Embarked']
)

```

```

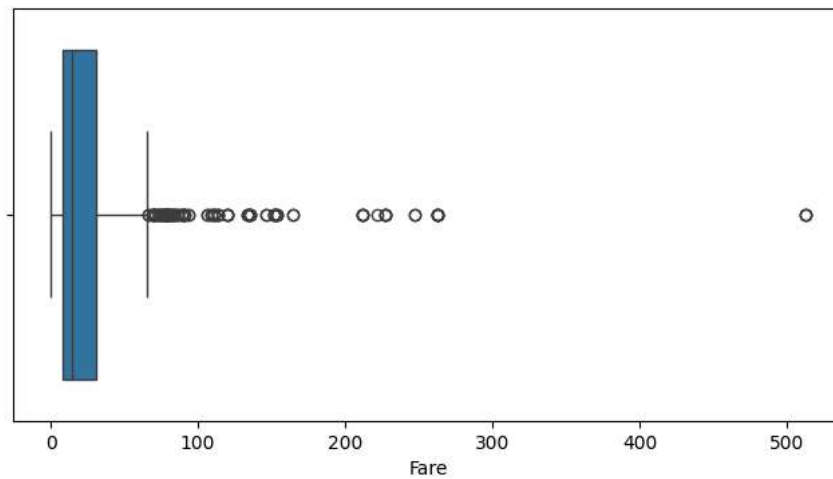
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(8,4))

sns.boxplot(
    x=df['Fare']
)

plt.show()

```



```
Q1 = df['Fare'].quantile(0.25)
Q3 = df['Fare'].quantile(0.75)
IQR = Q3 - Q1
```

```
lower = Q1 - 1.5*IQR
upper = Q3 + 1.5*IQR
```

```
df = df[
    (df['Fare'] >= lower) &
    (df['Fare'] <= upper)
]
```

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

df[['Age', 'Fare']] = scaler.fit_transform(
    df[['Age', 'Fare']]
)
```

```
df['FamilySize'] = (
    df['SibSp']
    + df['Parch']
    + 1
)
```

```
df['AgeGroup'] = pd.cut(
    df['Age'],
    bins=[-5, -1, 0, 1, 5],
    labels=[
        'Child',
        'Teen',
        'Adult',
        'Senior'
    ]
)
```

```
df.head()
```

|   | PassengerId | Survived | Pclass | Name                          | Sex | Age       | SibSp | Parch | Ticket              | Fare      | Embarked | FamilySize | AgeGroup |
|---|-------------|----------|--------|-------------------------------|-----|-----------|-------|-------|---------------------|-----------|----------|------------|----------|
| 0 | 1           | 0        | 3      | Braund,<br>Mr. Owen<br>Harris | 1   | -0.528321 | 1     | 0     | A/5 21171           | -0.779117 | 2        | 2          | Teen     |
| 2 | 3           | 1        | 3      | Heikkinen,<br>Miss.<br>Laina  | 0   | -0.215182 | 0     | 0     | STON/O2.<br>3101282 | -0.729373 | 2        | 1          | Teen     |
|   |             |          |        | Futrelle,                     |     |           |       |       |                     |           |          |            |          |

```
df.to_csv(
    f"{DATASET_PATH}/Titanic_Cleaned.csv",
```

```
        index=False  
    )
```

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